

AER1503 Project Definition

Optimizing transfer trajectories for Earth-Mars interplanetary orbital maneuvers

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I. PROJECT OBJECTIVE

The primary goal of this project is to develop a comprehensive Python-based software that identifies and optimizes interplanetary transfer trajectories between Earth and Mars. Using the OpenMDAO Python framework, the software will consider several design variables (i.e propulsion methods, transfer times, fuel minimization, etc) in order to recommend optimal missions to Mars. Since OpenMDAO is a multidisciplinary optimization (MDO) framework, the project will take advantage of its capabilities by optimizing several subsystems (i.e orbital mechanics, propulsion, thermal, and power) separately, which will then be combined into a larger global environment.

II. MOTIVATION

With the emergence of NASA's Artemis program and other global initiatives, the public's interest in space exploration has been renewed. Given that humanity has already been to the moon (and will likely get there again), the next feasible target for mankind is Mars. Despite its relatively close proximity to Earth, planning Martian missions is notoriously complex. Factors such as continually-changing planet positions, spacecraft design constraints, and mission length all contribute to the difficulty of space travel.

While a complete and independent solution to the problem of optimal interplanetary travel is beyond the scope of this project, we seek to address the matter by attempting to combine several subsystems within a flexible, open-source framework. Doing so will allow for a deeper understanding of what it will take to send humans to Mars, and might also provide insights that go beyond what is covered in the AER1503 course.

III. PLANNED STEPS

For the project to proceed as planned, there are several steps that will be taken. An outline of these steps is presented below:

- 1) Literature review of OpenMDAO, along with Chapter 10 content from the course AER1503. In particular, an optimal Hohmann transfer tutorial from OpenMDAO will be followed.
- 2) Determine exact design variables and decide on spacecraft parameters. These will serve as the basis for what the program will seek to optimize.

- 3) Creating a basic architecture of OpenMDAO systems and subsystems, which will be connected together as OpenMDAO inputs and outputs.
- 4) Expand on architecture using gradient-based optimization methods (still tentative).
- 5) Test a variety of initial design variables (i.e minimize delta-v, flight time) and see which ones are best optimized given various mission constraints.
- 6) Produce a final report that summarizes the project at large, along with conclusions drawn from the results of the transfer trajectory optimization.

The steps above form a general guideline for what the project is expected to consist of over the course of 4 months. As the project's scope and realistic outcomes become better defined, these steps may change.

IV. EXPECTED OUTCOMES

There are several goals that the project will hopefully meet by the end of the semester. In no particular order, they are:

- Produce a comprehensive, end-to-end Python program that will optimize the transfer trajectory of a Martian mission given a predetermined set of design variables.
- Provide a comparison of various types of transfer maneuvers (i.e Hohmann, low-thrust, gravity-assisted, etc) and determine what works best for several launch windows.
- Serve as a starting point for further research in trajectory optimization, which will be easy to adapt to a particular project's needs due to OpenMDAO's modular optimization approach.

Similar to the planned steps, these goals may be adjusted as the semester progresses.